

Clinical Study Report Synopsis: Study VLT-201 (ASCENT-HTN)

Sponsor: Northlake Biopharma. Investigational product: veltaprin (VLT-118), film-coated tablets for oral use, taken once daily in the morning.

Veltaprin is a selective aldosterone synthase inhibitor. It lowers aldosterone production by inhibiting the CYP11B2 enzyme, with low affinity for CYP11B1, the enzyme responsible for cortisol synthesis.

Indication studied: uncontrolled hypertension in adults who are already receiving three or more antihypertensive medicines.

Study title: A Phase 2b, randomized, double-blind, placebo-controlled, dose-ranging, parallel-group study of the efficacy and safety of veltaprin in adults with uncontrolled hypertension.

The study was conducted at 42 sites in 6 countries (Germany, Poland, Spain, Switzerland, the United Kingdom and the United States). The first participant was enrolled on 14 March 2022 and the last participant completed the last visit on 29 September 2023.

1. Objectives and study design

Primary objective: to evaluate the effect of veltaprin compared with placebo on the change from baseline in mean seated systolic blood pressure (SBP) at week 12. This change was the primary endpoint of the study.

Secondary endpoints: change from baseline in mean seated diastolic blood pressure (DBP) at week 12; proportion of participants with seated SBP below 130 mmHg at week 12; change in 24-hour ambulatory SBP in a substudy; safety and tolerability.

Design: after a 2-week single-blind placebo run-in period, eligible participants were randomized in a 1:1:1:1 ratio to placebo or veltaprin 2.5 mg, 5 mg or 10 mg once daily for 12 weeks, followed by a 4-week safety follow-up without study treatment. Randomization was stratified by baseline eGFR (below 60 versus 60 or above mL/min/1.73 m²) and by type 2 diabetes status.

Key inclusion criteria: age 18 to 80 years; mean seated SBP of at least 140 mmHg and below 180 mmHg while on stable doses of three or more antihypertensive medicines, one of them a diuretic, for at least 8 weeks.

Key exclusion criteria: serum potassium above 5.0 mmol/L at screening; eGFR below 45 mL/min/1.73 m²; myocardial infarction or stroke within the previous 6 months; New York Heart Association class III or IV heart failure; pregnancy or breastfeeding.

2. Study population

A total of 612 people were screened and 412 participants were randomized: 103 to placebo, 102 to veltaprin 2.5 mg, 104 to veltaprin 5 mg and 103 to veltaprin 10 mg. All randomized participants received at least one dose of study treatment.

Overall, 381 participants (92.5%) completed the 12-week treatment period. Thirty-one participants discontinued early: 12 because of adverse events, 9 withdrew consent, 6 were lost to follow-up and 4 discontinued for other reasons.

Baseline characteristics were balanced across groups. The mean age was 61.4 years and 46% of participants were female. Mean seated blood pressure at baseline was 152.8/88.1 mmHg. Type 2 diabetes was present in 38% of participants and the mean eGFR was 71 mL/min/1.73 m². Participants were taking a mean of 3.4 antihypertensive medicines at baseline.

3. Efficacy results

Primary endpoint. The least-squares mean change from baseline in seated SBP at week 12 was -4.1 mmHg with placebo, -9.8 mmHg with veltaprin 2.5 mg, -14.6 mmHg with 5 mg and -17.9 mmHg with 10 mg.

The placebo-adjusted differences were -5.7 mmHg for 2.5 mg (95% CI -9.2 to -2.2; $p=0.002$), -10.5 mmHg for 5 mg (95% CI -14.0 to -7.0; $p<0.001$) and -13.8 mmHg for 10 mg (95% CI -17.3 to -10.3; $p<0.001$). The analysis used a mixed model for repeated measures. The primary endpoint was met for all three doses.

Secondary endpoints. The mean change in seated DBP at week 12 was -1.9 mmHg with placebo and -4.2, -6.8 and -8.3 mmHg with veltaprin 2.5 mg, 5 mg and 10 mg.

The proportion of participants with seated SBP below 130 mmHg at week 12 was 14.6% with placebo, 27.5% with 2.5 mg, 41.3% with 5 mg and 49.5% with 10 mg.

In the ambulatory blood pressure substudy (168 participants), the placebo-adjusted change in 24-hour SBP with veltaprin 10 mg was -11.2 mmHg. Results were consistent across subgroups defined by baseline eGFR and diabetes status.

4. Safety results

Treatment-emergent adverse events were reported in 38.8% of participants on placebo and in 41.2%, 44.2% and 49.5% of participants on veltaprin 2.5 mg, 5 mg and 10 mg.

Hyperkalemia, defined as serum potassium above 5.5 mmol/L, was the adverse event of special interest. It occurred in 1 participant (1.0%) on placebo, 3 (2.9%) on 2.5 mg, 6 (5.8%) on 5 mg and 10 (9.7%) on 10 mg. Five participants discontinued treatment because of hyperkalemia: 1 in the 5 mg group and 4 in the 10 mg group.

Dizziness was reported in 2.9% of participants on placebo and in 3.9%, 5.8% and 6.8% on veltaprin 2.5 mg, 5 mg and 10 mg. Headache and fatigue occurred at similar rates in all groups.

Serious adverse events were reported in 9 participants (2.2%) overall. One serious event, hyperkalemia requiring hospitalization in the 10 mg group, was considered related to study treatment. No deaths occurred during the study.

Morning serum cortisol did not change in a clinically meaningful way and ACTH stimulation tests remained normal in all dose groups. The mean eGFR decreased by 3.2 mL/min/1.73 m² at week 12 in the 10 mg group and returned to baseline during follow-up.

5. Conclusions

Veltaprin given once daily for 12 weeks produced dose-dependent, clinically relevant reductions in systolic blood pressure in adults with uncontrolled hypertension on three or more medicines.

The 5 mg once-daily dose was selected for Phase 3. It delivered most of the blood pressure reduction seen with 10 mg while the rate of hyperkalemia was lower (5.8% versus 9.7%).

The Phase 3 program will include serum potassium monitoring at weeks 1, 4 and 12.

Limitations: treatment lasted 12 weeks, so durability of the effect is not known; 11% of participants were Black, which limits conclusions for this group; the study was not designed to assess cardiovascular outcomes.